



The Future of Tropical Rain Forests IDS 2935

5-2: Patterns of Tropical Biodiversity

*75% of the world's tigers are found in India, which has 58 official Tiger Reserves. The Bengal Tiger (*Panthera tigris tigris*) can reach 3 m from nose to tail and weight 280 kg (~617 lbs)*

1

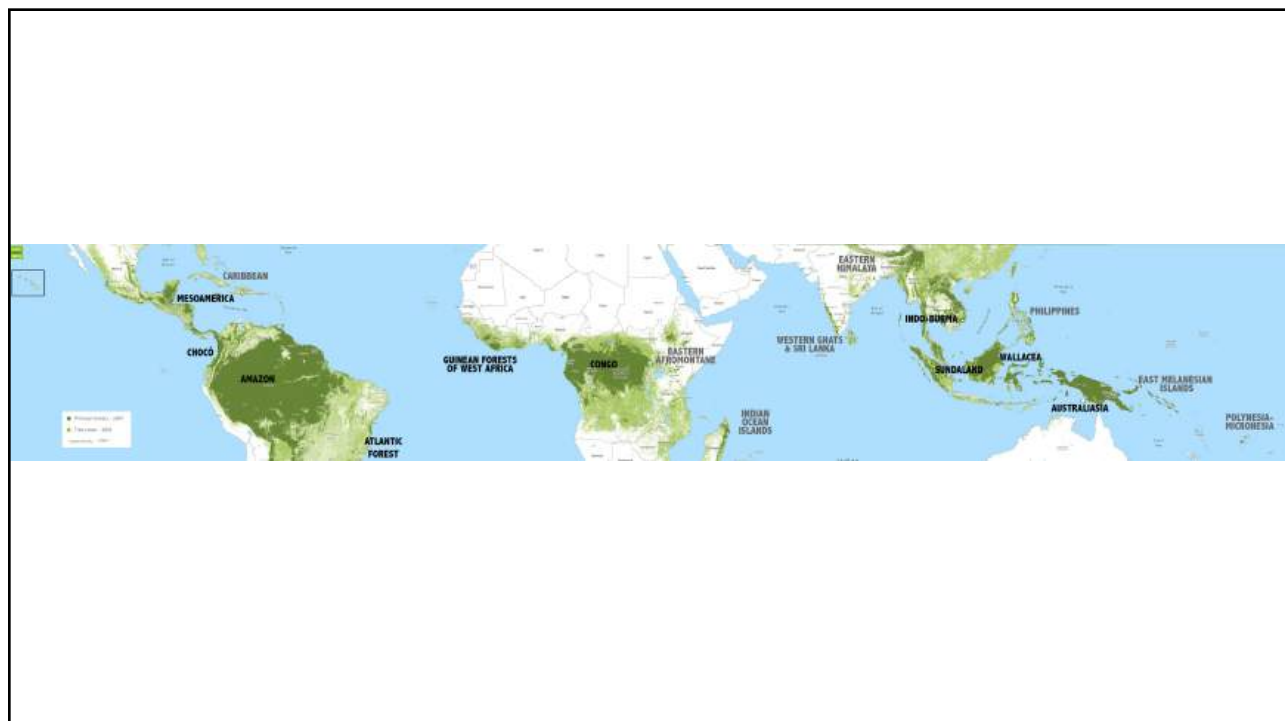




2



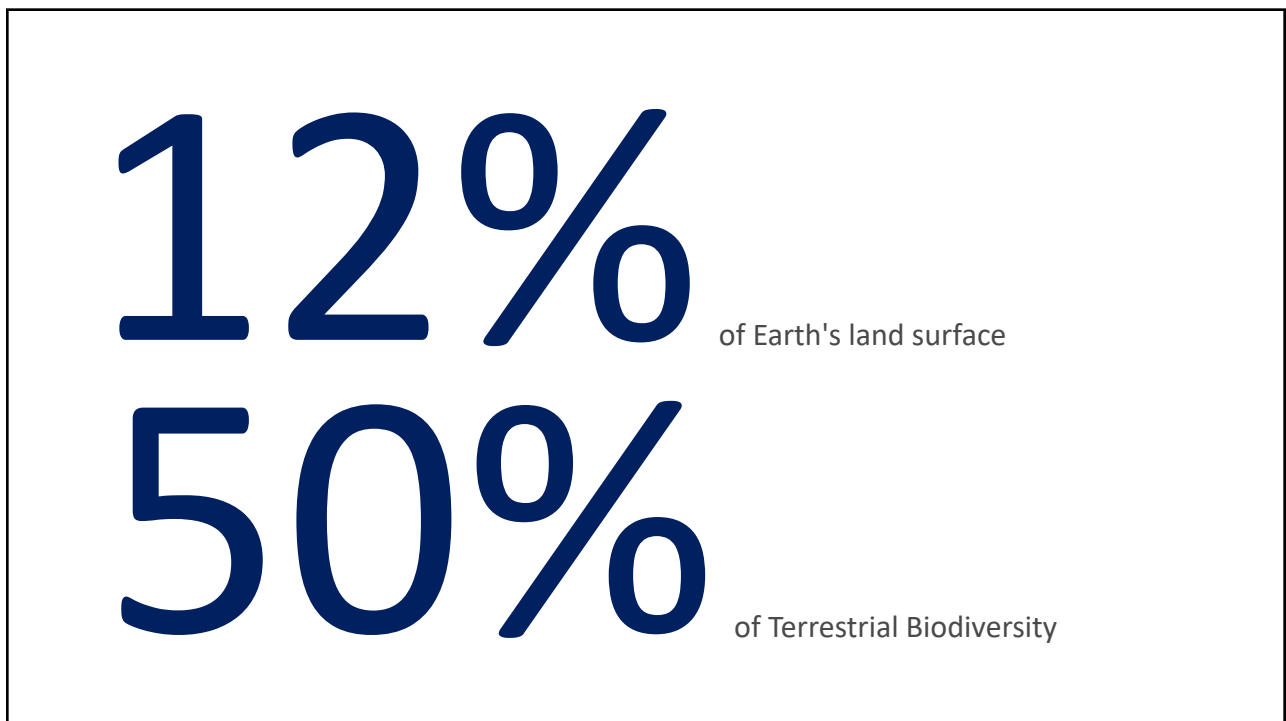
3



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6

How many species are there on Earth?

7



8.7 million $\pm$ 1.3 million SE
(2.2 million $\pm$ 0.18 million SE of these are marine)
Mora et al 2011 Plos Bio

8

Where are they?

9

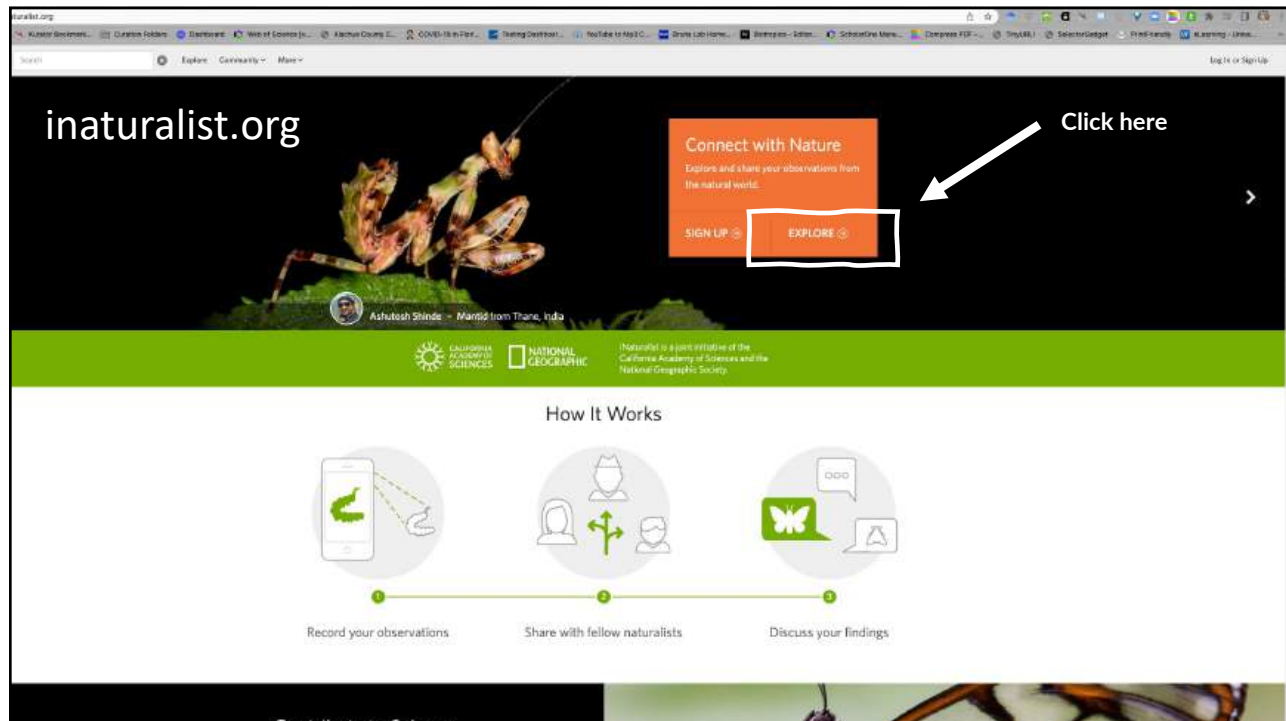


- 1: Juneau, AK, USA
- Winnipeg, CA
- 2: Buffalo, NY, USA
- 3: Cedar Rapids, IA , USA
- 4: Nashville, TN, USA
- El Paso, TX, USA
- 5: Gainesville, FL, USA
- Monterrey, Mexico
- 6: Cuernavaca, Mexico
- 7: Los Tuxtlas, Veracruz, Mexico
- 8: Gamboa, Panama
- 9: Medellin, Colombia
- 10: Manaus, Brazil
- Rio Branco, Brazil
- 11: Cuiabá, Brazil
- 12: Rio de Janeiro, Brazil
- Curitiba, Brazil
- 13: Rosario, Argentina
- Montevideo, Uruguay
- 14: Valdivia, Chile
- Torres del Paine, Chile
- 15: Punta Arenas , Chile

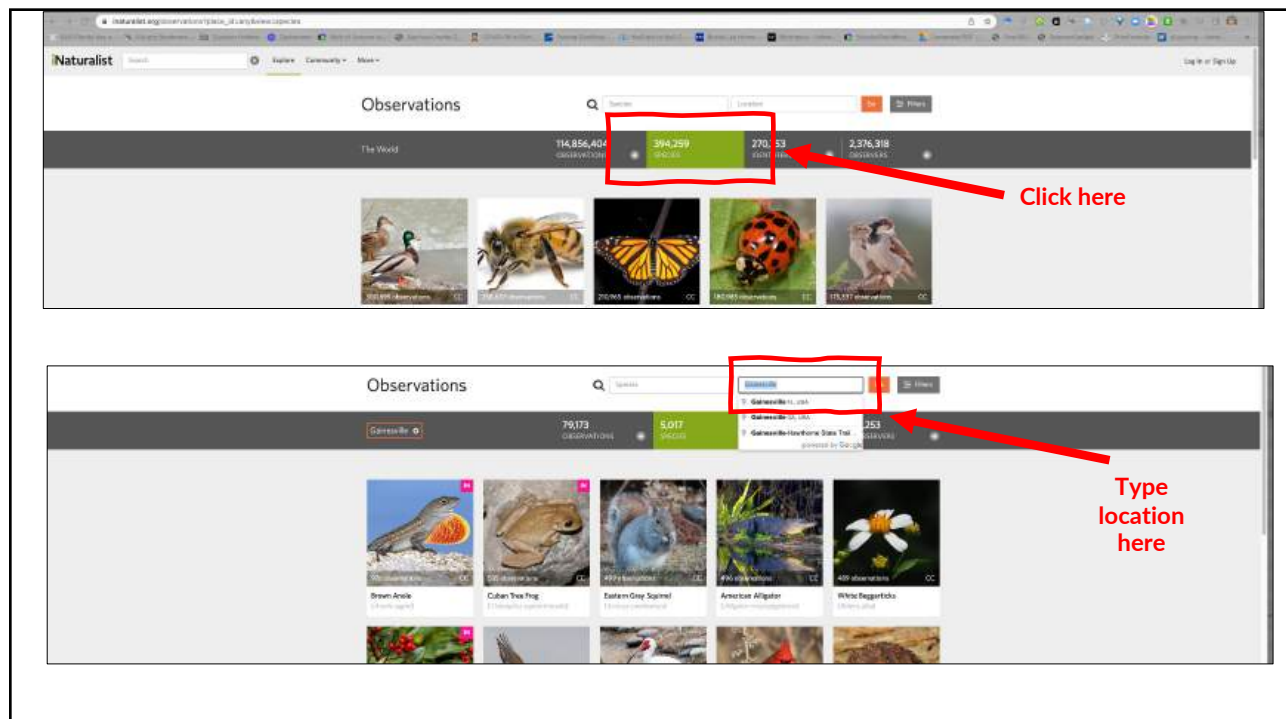


<https://www.inaturalist.org/>

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12

STEP 1
Record the following:
1) No. of Species
2) No. of Observers

(these are the TOTAL NUMBER for all the different types of species)

STEP 2
Click the "Filters" button

The screenshot shows the iNaturalist Observations page. At the top, there are four summary boxes: "24,370 OBSERVATIONS", "1,840 SPECIES", "1,806 IDENTIFIERS", and "1,808 OBSERVING". Red arrows point from the first two boxes to the "STEP 1" text. A red arrow points from the "Filters" button to the "STEP 2" text. Below the summary boxes is a grid of 20 species cards, each with a photo and name.

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STEP 3
Check
"Verifiable"
and
"Research Grade"
and
"Wild"

STEP 4
Click the "Bird" icon,
then "update search"

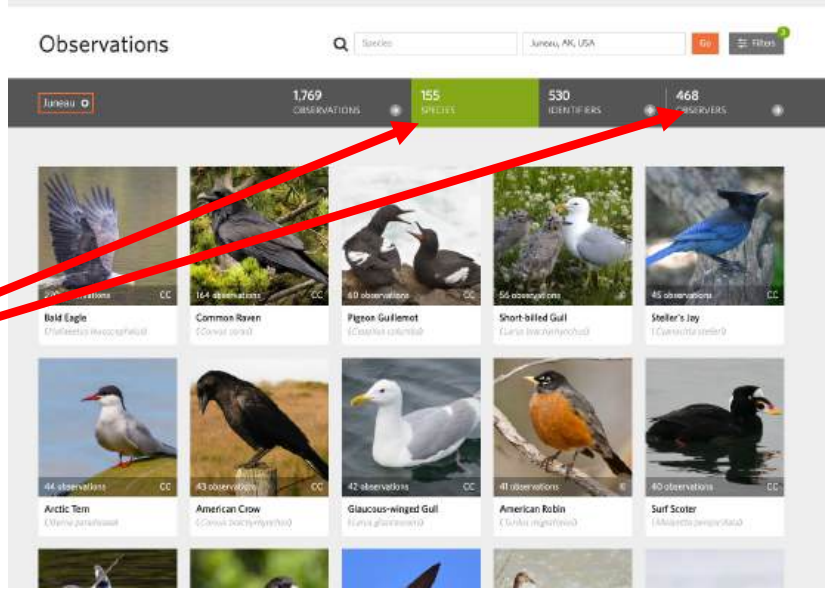
The screenshot shows the iNaturalist Observations page with the filters sidebar open. The "Show" section has "Wild", "Verifiable", and "Research Grade" checked. The "Date Observed" section has "Any" selected. A blue arrow points from the "Bird" icon in the "Categories" section to the "STEP 4" text. A red arrow points from the "Verifiable" checkbox to the "STEP 3" text. The main grid shows 10 species cards, including Humpback Whale, Devil's Club, Bald Eagle, Nootka Lupine, Western Skunk Cabbage, Western Bunchberry, Salmonberry, Dwarf Fireweed, Geatsbeard, and North American Porcupine.

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STEP 5
Record :

- 1) Number of Bird Species
- 2) Number of Observers

Why not check these amazing animals while you're here?



Observations

1769 OBSERVATIONS 155 SPECIES 530 IDENTIFIERS 468 OBSERVERS

Bald Eagle (Haliaeetus leucocephalus) 164 observations

Common Raven (Corvus corax) 40 observations

Pigeon Guillemot (Ceophaps columba) 56 observations

Short-billed Gull (Larus brevirostris) 45 observations

Stellar's Jay (Cyanocitta stelleri) 44 observations

Arctic Tern (Stercorarius pomarinus) 43 observations

American Crow (Corvus brachyrhynchos) 42 observations

Glaucous-winged Gull (Larus glaucescens) 41 observations

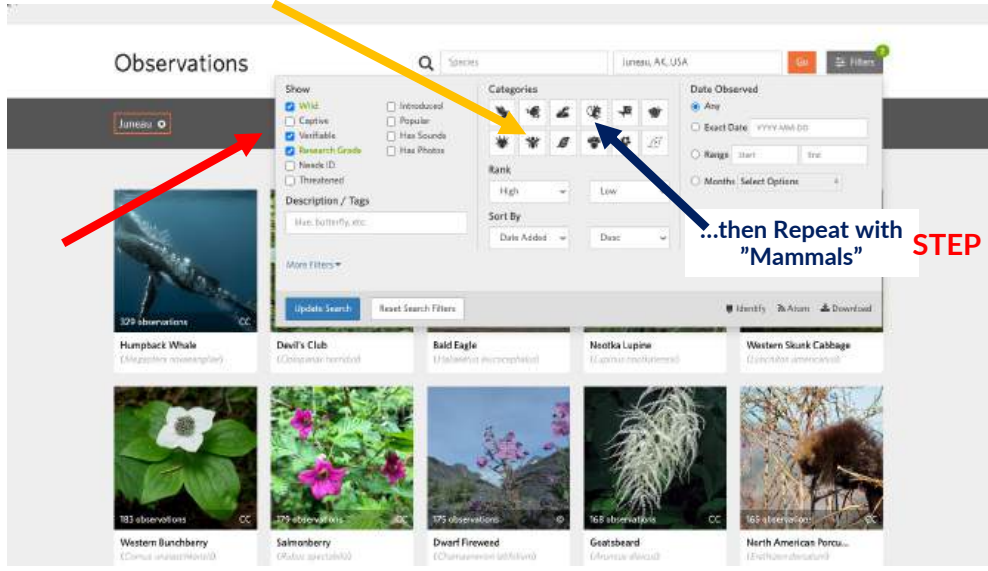
American Robin (Erubus migratorius) 40 observations

Surf Scoter (Melanerpes persimilis)

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STEP 6 Uncheck "Birds" and repeat with "Insects" and collect the same information (be sure to Keep "Verifiable", "Research Grade", and "Wild" Checked.)

...then Repeat with "Mammals" **STEP 7**



Observations

1769 OBSERVATIONS 155 SPECIES 530 IDENTIFIERS 468 OBSERVERS

Wild ☒ Captive ☐ Verifiable ☒ Research Grade ☒ New ID ☐ Threatened ☐ Introduced ☐ Popular ☐ Has Sounds ☐ Has Photos ☐

Description / Tags

Map, butterfly, etc.

More Filters

Update Search Reset Search Filters

Categories

Rank

Sort By

Date Added Desc

Date Observed

Any

Exact Date YYYY-MM-DD

Range Start End

Months Select Options 4

Humpback Whale (Megaptera novaeangliae) 129 observations

Devil's Club (Opodisma nortoni) 123 observations

Bald Eagle (Haliaeetus leucocephalus) 175 observations

Neotoma Lupine (Lupinus neotomae) 163 observations

Western Skunk Cabbage (Lysichiton americanus) 160 observations

Western Bunchberry (Cornus vancouveriana) 123 observations

Salmonberry (Rubus spectabilis) 175 observations

Dwarf Fireweed (Chamaenerion latifolium) 175 observations

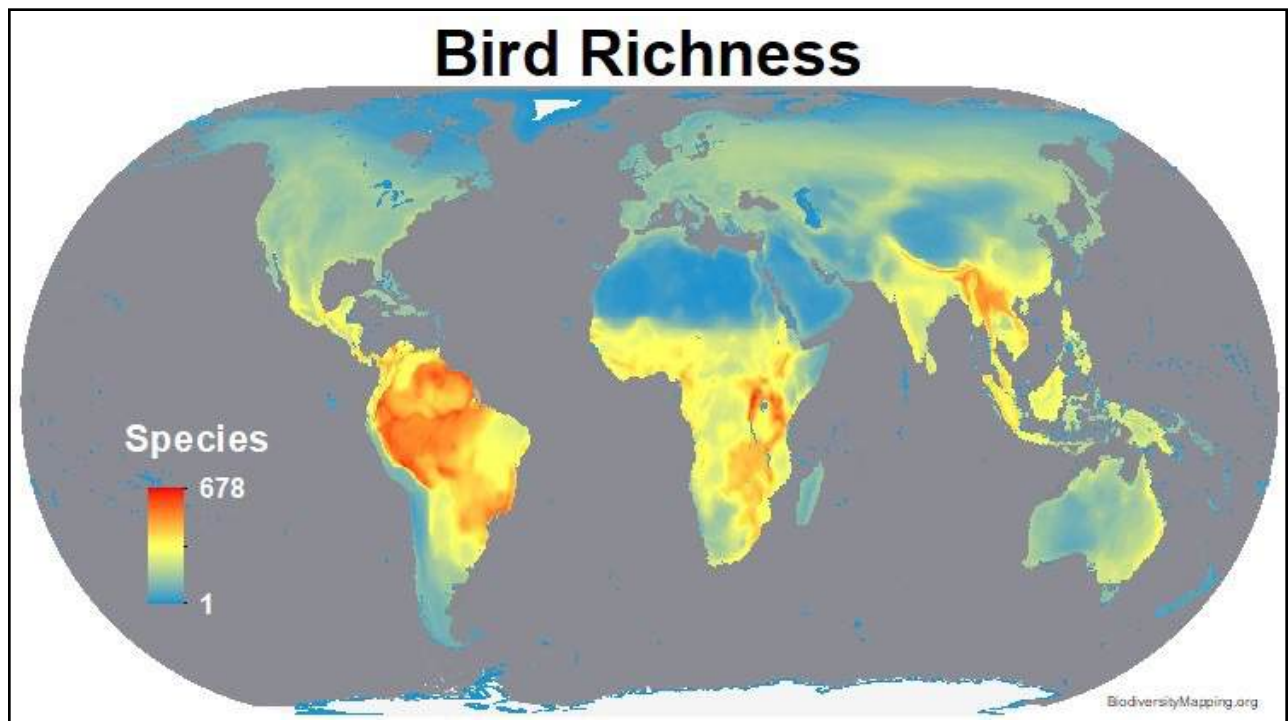
Geatsbeard (Lonicera chrysantha) 163 observations

North American Porcupine (Erethizon americanum) 160 observations

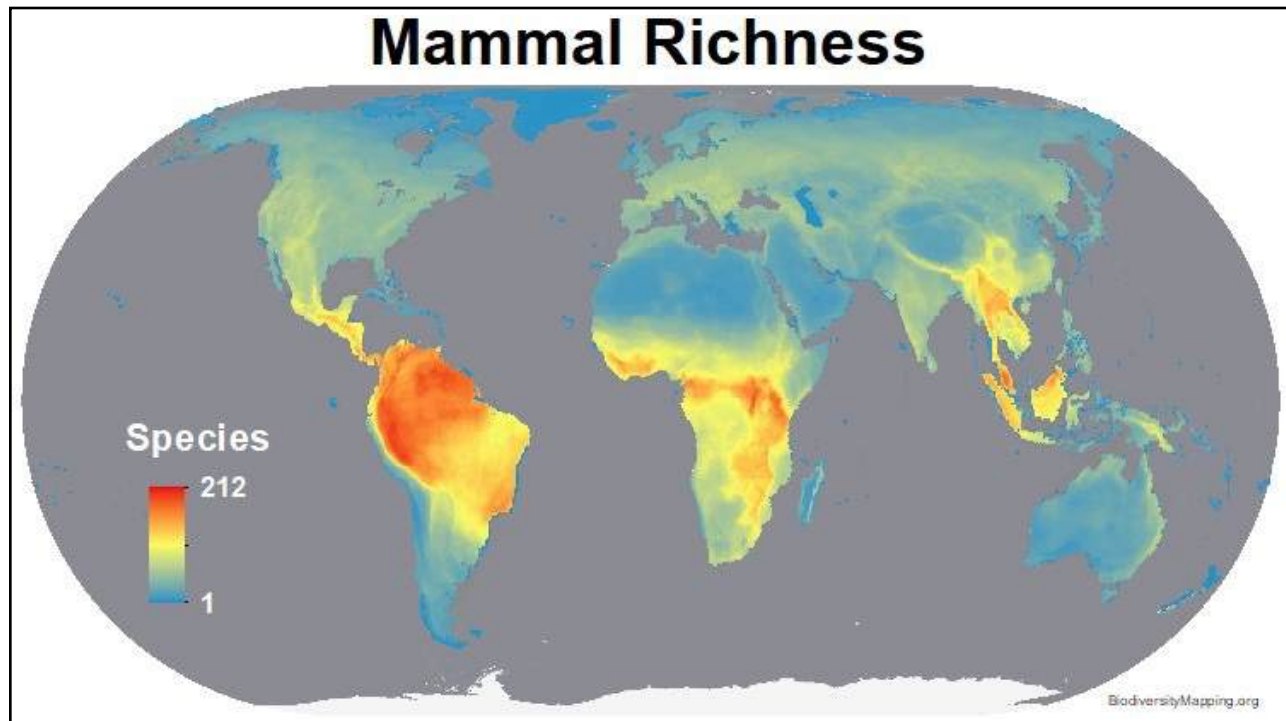
16



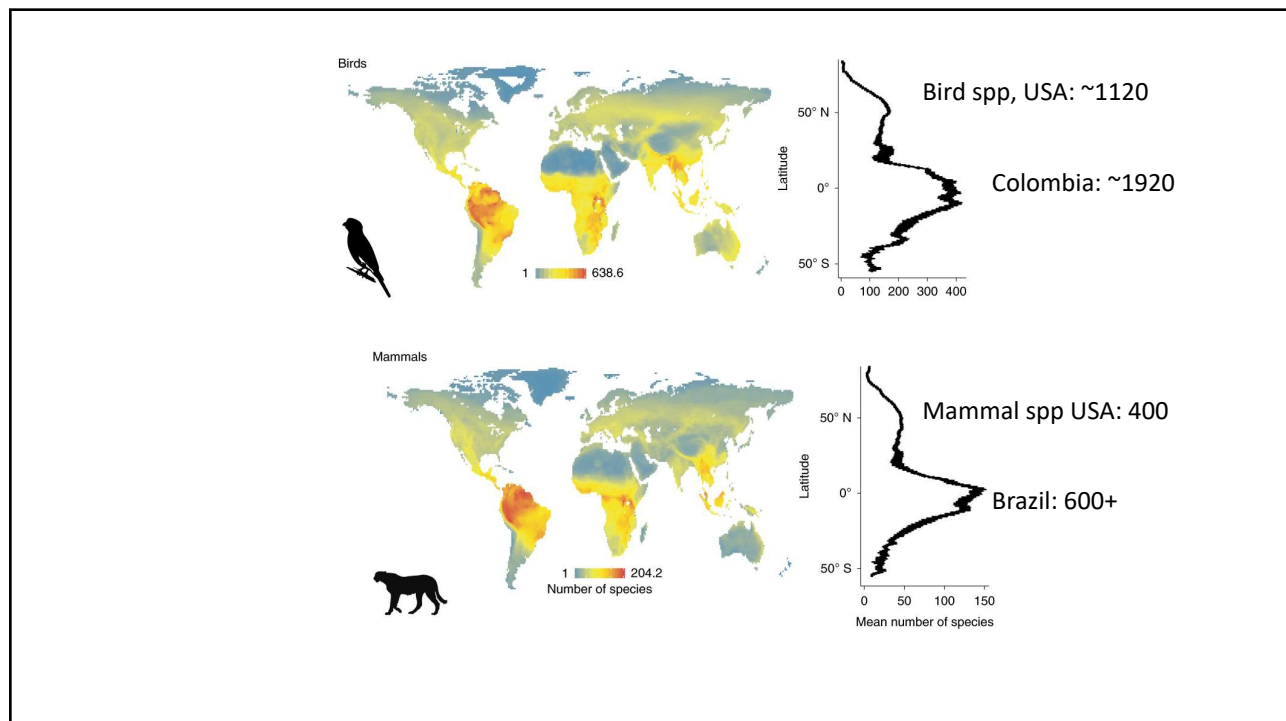
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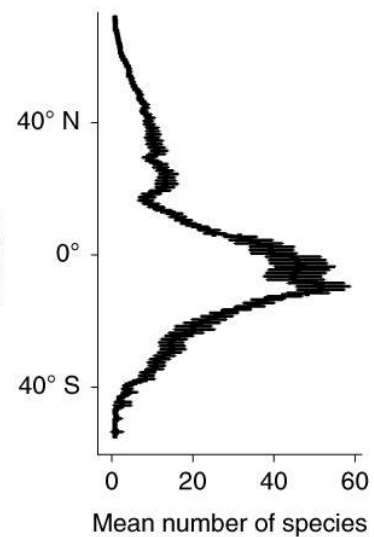
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Fundamental Biodiversity Pattern #2:

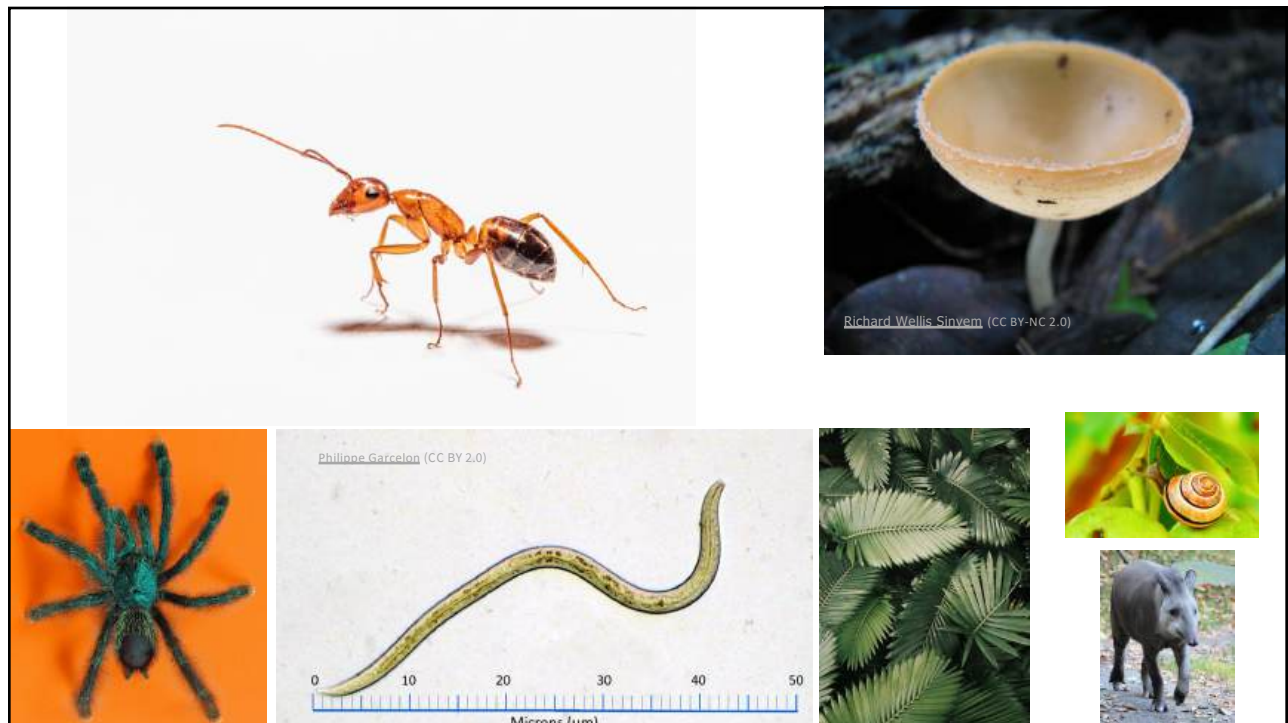
Latitudinal Diversity Gradient



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What are they?

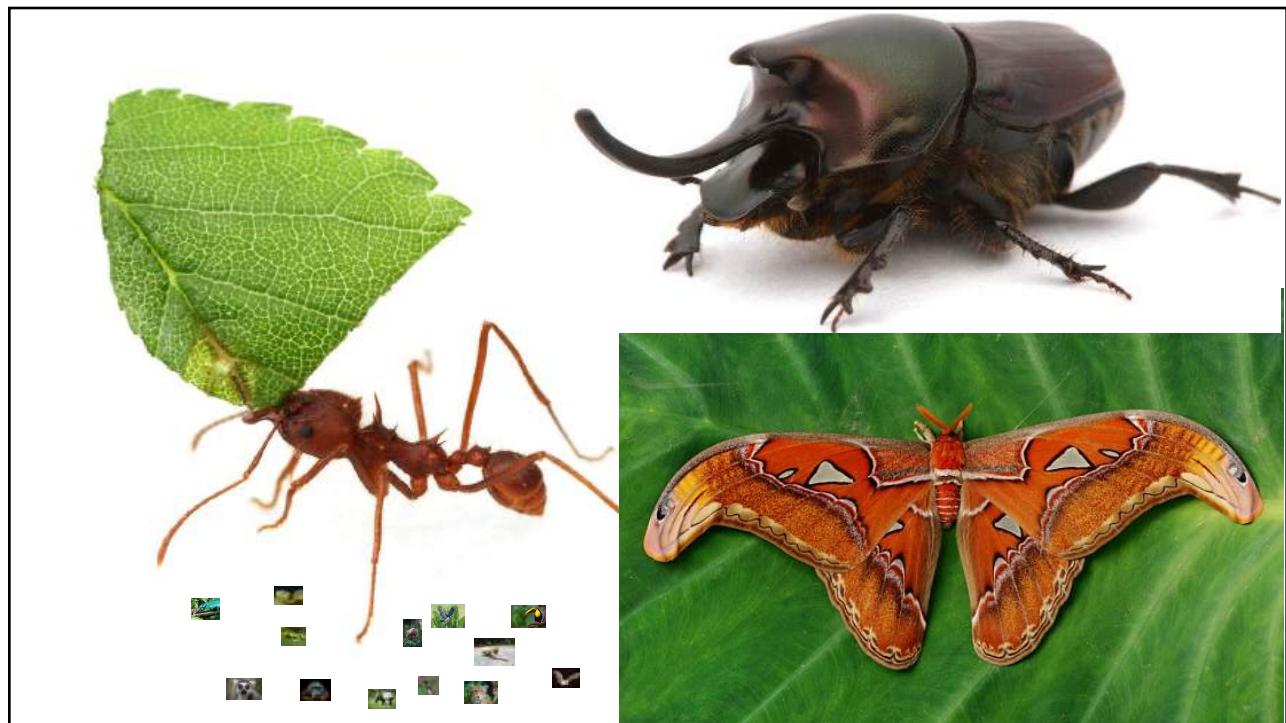
22



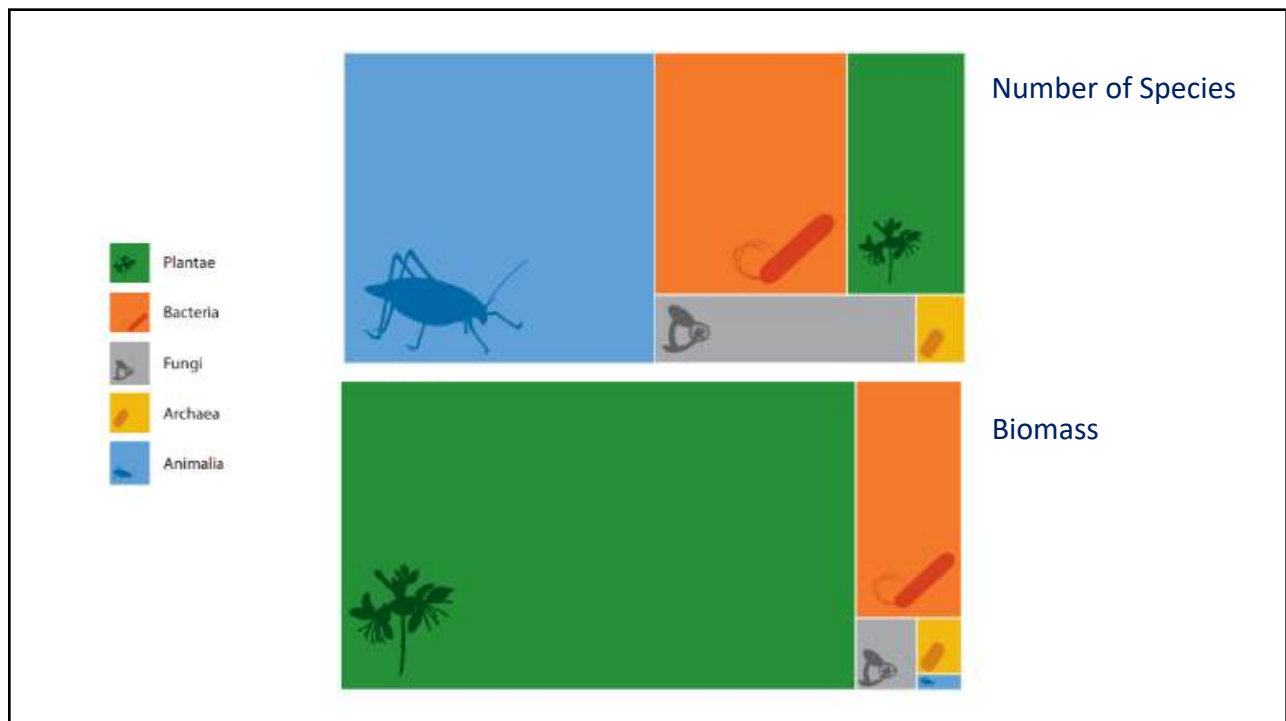
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24



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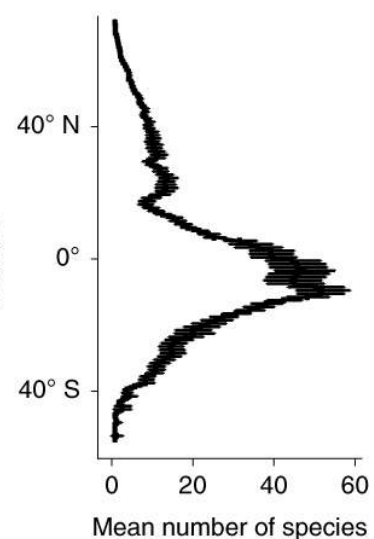
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We have now inferred two
Fundamental Patterns of Biodiversity

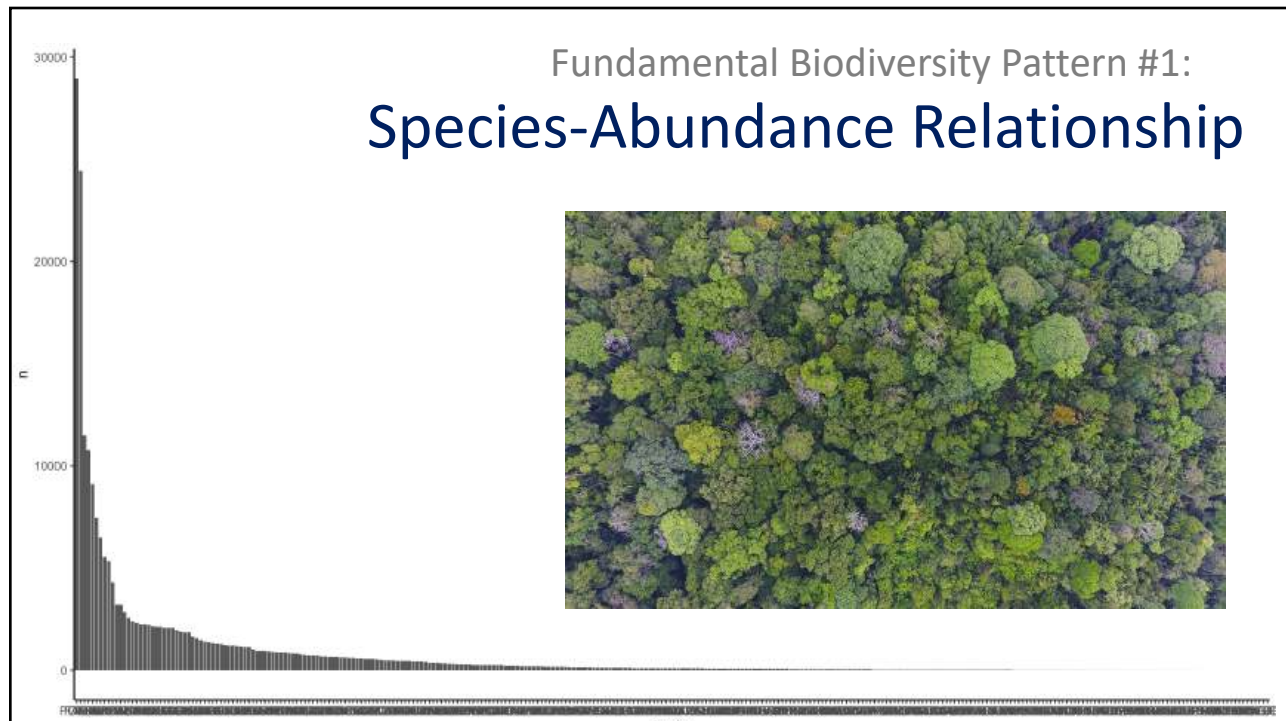
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Fundamental Biodiversity Pattern #2:

Latitudinal Diversity Gradient



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There are approx. 37,900
named tropical tree species
in the tropics and....

...there are an estimated
800 billion tropical trees
with trunk diameters of at
least 10 cm.



50% of these 800 billion trees belong to only 1,053 species.

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